

Multitemporal and Multimodal Cross-Region Fine-Tuning: MultiSenGE and MultiSenNA Applications

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ABSTRACT

This letter introduces the MultiSenNA dataset, a new large-scale, multitemporal, and multimodal semantic segmentation and classification resource for the *Nouvelle-Aquitaine* region in France. This dataset is an additional resource to the new collection of Artificial Intelligence benchmark datasets (AI4LCC). Building on the existing MultiSenGE dataset, MultiSenNA aims to enhance land use and land cover (LULC) classification through deep learning, especially focusing on urban fabrics and their role in climate models. This work examines the effectiveness of fine-tuning pre-trained models for LULC mapping, hypothesizing that this approach not only yields better results than models with random initialization but also reduces the carbon impact of the training process. The study uses an existing deep learning network, *ConvLSTM+Inception-S1S2*, applying it to the MultiSenNA and MultiSenGE datasets. Key findings include the demonstration of fine-tuning's robust performance on both full and small datasets, maintaining high accuracy and F1-Scores while significantly lowering carbon emissions and execution time. This research highlights the necessity of transitioning towards large-scale foundational models that can be fine-tuned for specific remote sensing applications. This approach has the potential to mitigate the considerable carbon emissions associated with current training processes, which encompass data acquisition, pre-processing, and model training.

KEYWORDS

fine-tuning, land use land cover, deep learning, semantic segmentation, MultiSenNA, MultiSenGE, benchmark dataset

1. Introduction

The advent of satellite imagery and advancements in remote sensing have revolutionized our understanding of the Earth's surface. The increasing availability of high-resolution, multitemporal and multimodal satellite data has opened new avenues for land use and land cover (LULC) classification, especially by using deep learning with semantic segmentation applications (Vali, Comai, and Matteucci 2020).

In recent years, global climate change, closely linked to anthropogenic activities, has continued to grow and significantly impacts natural and human-modified landscapes. In this context, precise and generic LULC data is needed for effective LULC monitoring. This is particularly true for urban fabrics, which need to be comprehensive

and generalizable for the majority of global cities and are integrated in many climate models for urban climate modeling and prediction as they characterize Local Climate Zones (LCZ).

With the advent of deep learning methods, the demand for diverse datasets has continually increased. However, most of these datasets lack diversity in urban fabric classes (Stanimirova et al. 2023), which are essential for numerous thematic applications such as Local Climate Zone (LCZ) characterization and urban climate modeling. Furthermore, the characteristics of these urban areas change rapidly and vary greatly depending on the location of the cities and are difficult to generalize across multiple territories, highlighting the need for location-specific data in LULC research. The MultiSenNA dataset was specifically developed to address the lack of large-scale, multitemporal and multimodal datasets representing temperate and western urban landscapes with detailed urban fabric classes. It complements existing resources such as MultiSenGE, enabling the study of transfer learning between different geographic contexts, and facilitating the development of foundation models adapted to diverse urban forms.

Despite the remarkable deep learning capabilities in LULC mapping, its application faces significant challenges, most notably the extensive time and computational resources required for training models (Strubell, Ganesh, and McCallum 2019). These requirements often translate into substantial carbon footprints, a growing concern in an environmentally conscious era (Verdecchia, Sallou, and Cruz 2023). To address these challenges, the field is shifting towards transfer learning and especially fine-tuning approaches (Gadiraju and Vatsavai 2023), with recently the development of the foundation model (Jakubik et al. 2023). Fine-tuning pre-trained models on specific datasets significantly shortens training time and limits carbon impact, offering a more sustainable and efficient pathway for leveraging deep learning in Earth observation.

In this letter, we propose a new large-scale multitemporal and multimodal benchmark dataset for the largest region in France, *Nouvelle-Aquitaine*. This dataset is part of the AI4LCC collection and complements MultiSenGE (Wenger et al. 2022b) (developed over the *Grand-Est* region in France). It enables users to perform various transfer learning approaches in multitemporal and multimodal satellite imagery for semantic segmentation. Using the existing multimodal and multitemporal deep learning network *ConvLSTM+Inception-S1S2* (Wenger et al. 2022a), this letter makes three key contributions: (1) the creation of the MultiSenNA dataset, a new large-scale multimodal and multitemporal benchmark dataset that complements the MultiSenGE dataset, (2) the proposal of a baseline approach for leveraging MultiSenGE and MultiSenNA, specifically investigating the hypothesis that fine-tuning pre-trained networks outperforms training from scratch for LULC mapping, and (3) an analysis of how reducing the size of the fine-tuned dataset can significantly lower the carbon impact of models with minimal accuracy loss, paving the way for more sustainable AI practices and large-scale foundation model developments in remote sensing. This perspective, still largely underexplored in remote sensing AI, highlights the need for more sustainable model development practices.

2. Datasets and Preprocessing

In this section, we will present the dataset used for this transfer learning approach (section 2.1) and the novel dataset created, MultiSenNA (section 2.2).

2.1. *MultiSenGE*

MultiSenGE (Wenger et al. 2022b) is a land use land cover benchmark dataset covering the entire *Grand-Est* region in France. This dataset takes as input as ground reference a vector database produced by photo-interpretation of aerial imagery, Sentinel-1 and Sentinel-2 time series with a restrictive cloud cover of 10% for Sentinel-2. At its final stage, MultiSenGE contains 8,157 patches with a size of 256×256 pixels.

2.2. *MultiSenNA*

MultiSenNA (for Multi-temporal/modal Sentinel-1/2 *Nouvelle-Aquitaine*) is a new dataset developed over the largest region in France, *Nouvelle-Aquitaine* which represents $84,061 \text{ km}^2$ (Figure 1). It follows on from the MultiSenGE (Wenger et al. 2022b) dataset, which was initially developed for the *Grand-Est* region of France.

Sentinel-2 imagery, containing 13 spectral bands with 10m to 60m spatial resolutions, was used for 2020. A Satellite Image Time Series (SITS) of 22 tiles, depicted in Figure 1, was downloaded from Theia land data services. These Level-2A (L2A) images, atmospherically corrected, were selected for minimal cloud cover (less than 10%) using the 'theia download' script. The MultiSenNA dataset incorporated the 10's high resolution bands: four at 10m resolution (B2, B3, B4, B8) and six at 20m resolution (B5, B6, B7, B8A, B11, B12). Bands at 20m have been resampled to 10m.

Sentinel-1, equipped with C-band Synthetic Aperture Radar (SAR), offers day-night, all-weather imagery. Its data, in dual polarization, are available as GRD and SLC products. For this dataset, 2020 SAR images from both orbits were processed using the *S1-Tiling* chain, developed by *Centre National des tudes Spatiales* (CNES), in Ground Range Detected (GRD). This pipeline includes Sentinel-1 data retrieval via the Earth Observation Data Access Gateway (EODAG) library, SAR data segmentation to align with the Sentinel-2 grid, orthorectification, and a multitemporal filter application for speckle reduction, preserving spatial information integrity (Filippini 2019).

Patches with the size of 256×256 pixels has been cut over both Sentinel-1 and Sentinel-2 time series and also over the reference data. We also constructed GeoJSON files containing the same information as the first dataset. The temporal distribution of this dataset for both sensors is reported in figure 3. It is worth noting that the temporal distribution of Sentinel-2 data is heterogeneous, with relatively few images available for October and December 2020. In its final version, MultiSenNA contains 12,261 multitemporal and multimodal patches and occupies approximately 500 GB of storage.

For MultiSenNA, we used a vector ground reference data developed by a visual interpretation of aerial photographs for 2019/2020. This reference data includes a four-level hierarchical nomenclature where 3 levels are based on Corine Land Cover. We applied the same preprocessing steps as those used in the production of MultiSenGE (detailed in Wenger et al. (2022b)). The level 4 has been reworked to match the 14 classes. However, due to the coastal localization of the *Nouvelle-Aquitaine* region, we decided to add one more class to this dataset compared to MultiSenGE, **beaches**, **sand** (Figure 2). This reference data has also been rasterized for the future treatments. Like most land use land cover datasets, this one is also highly imbalanced. For example, urban fabric classes represent around 5% of the total land cover surfaces and the new class beaches, sand is the less represented one with 0.11% (Figure 2). To facilitate

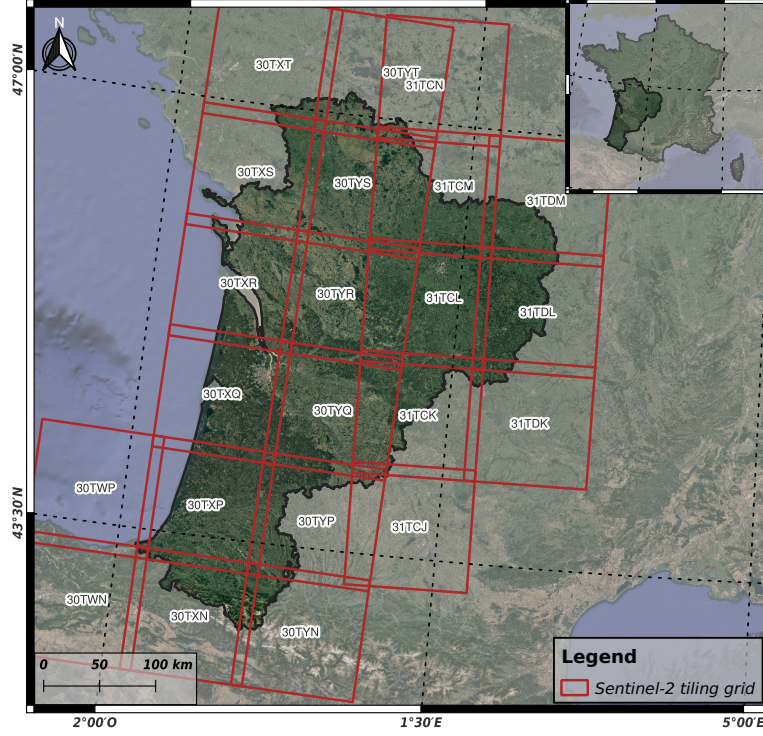


Figure 1. Nouvelle-Aquitaine region (France) where MultiSenNA was developed. Coordinates are in WGS84.

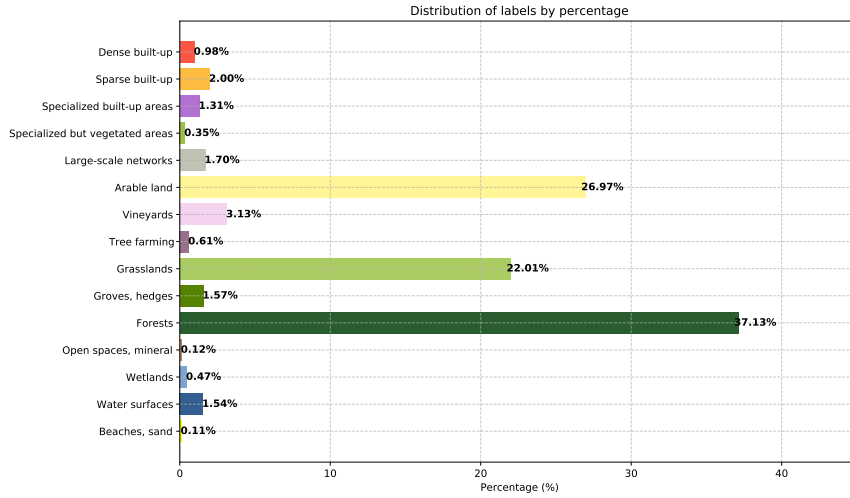


Figure 2. Ground reference classes distribution over MultiSenNA dataset.

access to the dataset, we provide several Python scripts available on GitHub¹.

3. Transfer learning

In traditional machine learning, models are initialized and trained from scratch for each specific task, a process that is both time and resource-intensive, especially for

¹<https://github.com/r-wenger/MultiSenGE-NA-Tools>

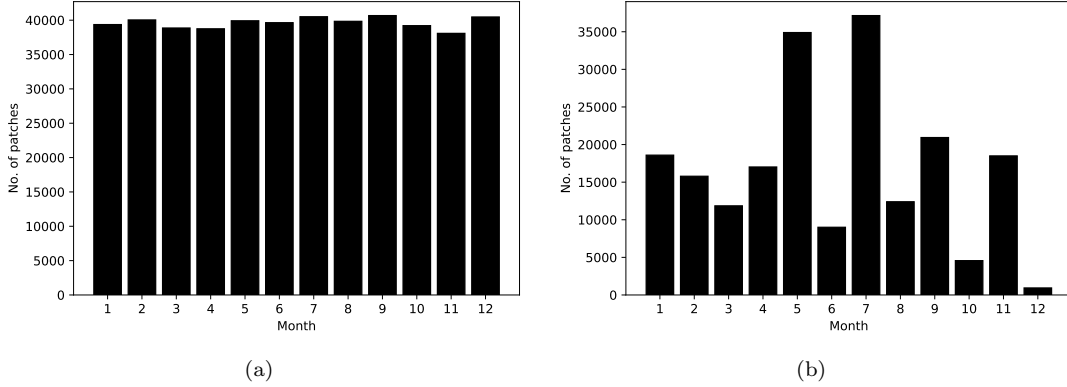


Figure 3. Temporal distribution of the patches for Sentinel-1 (a) and Sentinel-2 (b) over the entire MultiSenNA dataset.

large neural networks, often needing multiple Graphics Processing Units (GPUs) for optimal classification performance. Transfer learning, however, presents substantial efficiency by allowing the reuse of knowledge from one task to another. In essence, the lower layers of a neural network, responsible for extracting generic features such as edges and textures in image processing, can be applied to new tasks with minimal adjustment. The adaptation primarily involves retraining the network’s higher layers to accommodate the unique characteristics of the new task, thereby reducing the computational burden and expediting the training process.

Although the distance between the *Grand-Est* and *Nouvelle-Aquitaine* regions remains small, these regions exhibit substantial domain differences due to variations in land cover types, climatic conditions, and agricultural practices. For instance, Nouvelle-Aquitaine is characterized by extensive vineyards and pine forests, whereas Grand-Est is dominated by temperate deciduous forests and urbanized landscapes. These differences translate into distinct spectral and temporal patterns in remote sensing data, especially for Sentinel-1 and Sentinel-2 imagery. The MultiSenNA dataset supports training, fine-tuning, and evaluating deep learning models designed for multitemporal and multimodal remote sensing data. Its substantial size and detailed class taxonomy also position it as a promising candidate for developing large-scale foundational models (also known as *giga models*) that can effectively characterize urban areas, biodiversity, and support sustainable land management.

Moreover, the multitemporal and multimodal nature of the dataset is important for building models that can generalize well across seasons, climatic variations, and land cover types, especially for applications targeting dynamic environments. Such characteristics are particularly important for advancing self-supervised learning approaches in Earth Observation, where pretraining on large, diverse, and temporally rich datasets like MultiSenNA helps reduce the reliance on expensive labeled data. This represents a step toward enabling robust and transferable models applicable to underrepresented regions, including many countries in the Global South, where annotated LULC data is scarce or unavailable.

3.1. Deep learning model

To perform this fine-tuning approach on these two LULC datasets, we used an existing deep learning model, *ConvLSTM+Inception-S1S2* (Wenger et al. 2022a), which

combined a spatio-spectral extractor (*Inception* module) over the concatenation of the first two dates of the series and a spatio-temporal extractor (*ConvLSTM* module) over the n Sentinel-1 and Sentinel-2 dates (in our case, four dates). The entire model can be summarized as follows:

$$\begin{aligned}
X_{S2}, X_{S1}, X_{\text{stack}} &: \text{Inputs for S2, S1, and stacked data} \\
X_{S2_out, -} &= \text{ConvLSTM2D}(X_{S2}, \text{initial_state}) \\
X_{S1_out, -} &= \text{ConvLSTM2D}(X_{S1}, \text{initial_state}) \\
X_{\text{stack_out}} &= \text{InceptionModule}(X_{\text{stack}}) \\
X_{\text{concat}} &= [X_{S2_out}, X_{S1_out}, X_{\text{stack_out}}] \\
\text{Output} &= \text{U-Net}(X_{\text{concat}})
\end{aligned} \tag{1}$$

where ConvLSTM2D means *Convolutional Long Short-Term Memory 2D*, S1 means Sentinel-1, S2 means Sentinel-2.

The weighted categorical cross-entropy loss function is used to handle the imbalance in class distribution. The weights are typically inversely proportional to the class frequencies in the training set (Audebert, Le Saux, and Lefèvre 2018; Wenger et al. 2022a). This approach helps focus the model’s learning more on the underrepresented classes, thereby addressing the issue of class imbalance.

3.2. *Fine-tuning approach*

Fine-tuning is essential in transfer learning, adapting pretrained neural networks efficiently to new datasets (Lu et al. 2024). Mathematically, fine-tuning updates pre-trained parameters ($\Theta_{\text{pre-trained}}$) on a new dataset (D_{new}) via gradient descent:

$$\Theta_{\text{fine-tuned}} = \Theta_{\text{pre-trained}} - \eta \nabla_{\Theta} L(\Theta_{\text{pre-trained}}, D_{\text{new}}) \tag{2}$$

where η is the learning rate, and $\nabla_{\Theta} L$ the gradient of the loss function L . We specifically investigate selective fine-tuning by freezing encoder layers of our *ConvLSTM+Inception-S1S2* model, updating only decoder layers. This strategy leverages previously learned general features, to efficiently adapt higher-level representations to the new dataset.

3.3. *Data augmentation*

Our augmentation techniques are selectively applied to both Sentinel-1 and Sentinel-2 image patches and their corresponding labels. The key transformations include:

- **Rotation:** Images are rotated by a random degree (90 to 270). This introduces rotational variance, helping the model generalize better to different orientations.
- **Flipping:** Both horizontal and vertical flips are employed. This randomness ensures that the model is not biased towards a specific directionality in the imagery.

We chose these two augmentation techniques to preserve the nature of Sentinel satellite imagery, which is rarely blurred or zoomed. Each image-label pair undergoes

these transformations randomly, ensuring a diverse and representative dataset. Additionally, both methods limit spatial autocorrelation, which is not insignificant when working with geographic and spatialized data.

3.4. *Technical implementation*

PyTorch version 2.0.1 has been used for these experiments. For the models, we selected an Adam optimizer with a learning rate of 10^{-3} . A training duration of 50 epochs was set after several tests. All Python scripts were executed on a GPU cluster, equipped with three RTX 6000 GPUs, each boasting 24 GigaByte (GB) of Video Random Access Memory (VRAM), cumulatively amounting to 72 GB. This configuration enabled the utilization of a batch size of 16 for each experiment detailed in the paper. We monitored the carbon impact of all of our models using *CodeCarbon* library (Schmidt et al. 2021).

4. Experiments

To match the previous work using *ConvLSTM+Inception-SIS2*, we selected four temporal depths for MultiSenNA as well. Due to the imbalanced temporal distribution of Sentinel-2 patches (Figure 3 B), we selected the first patch available for May, July, September, and November. For MultiSenGE, July, August, September, and November have been selected according to (Wenger et al. 2022a). We selected 10 LULC classes (described in (Wenger et al. 2022a)) for both datasets to perform this research, 5 urban fabric classes and 5 natural classes (we regrouped some of the less represented classes).

In this study, we developed a test protocol that combines several fine-tuning approaches across both datasets. In the first step, we compared the geographical stratification with random patches selection. Both approaches were applied to each datasets using random weights initialization (Test 1 to 4 in Table 1). Then we used pretrained weights to fine-tune all layers of the network to train the network over MultiSenNA (using the network trained over MultiSenGE) and over MultiSenGE (using the network trained over MultiSenNA) to explore fine-tuning for LULC classification (Tests 5 and 6 in Table 1). As a second step, we froze the encoder part of the U-Net to train over MultiSenNA using MultiSenGE weights (Test 7 in Table 1). Finally, we reduced the dataset size by a factor of four to force the network to train on a smaller dataset, in order to assess its generalization capacity in areas with limited ground reference data (Test 8 in Table 1).

5. Results

The performance of Ensemble Transfer Learning models MultiSenNA and MultiSenGE in land use and land cover (LULC) classification was evaluated through a series of experiments, as detailed in Table 2.

For MultiSenNA, the highest accuracy was observed in test 5 (81.90%), where the model underwent fine-tuning with weights pretrained on MultiSenGE. This approach also yielded the highest weighted F1-Score (83.17%) for MultiSenNA, underscoring the effectiveness of transfer learning and fine-tuning in enhancing model performance across various classes. MultiSenGE demonstrated a similar trend, with its best performance in test 6 (accuracy of 85.55% and a weighted F1-Score of 86.47%), where

Table 1. Detailed summary of experiments in LULC classification.

Test No.	Approach	Dataset	Weight Initialization	Remarks
1	Geographic Stratification	MultiSenNA	Random	Initial test with geographic stratification
2	Random Selection	MultiSenNA	Random	Initial test with random patch selection
3	Geographic Stratification	MultiSenGE	Random	-
4	Random Selection	MultiSenGE	Random	-
5	Fine-tuning	MultiSenNA	Pretrained (MultiSenGE)	Fine-tuning all layers for LULC (weights from test 4) - (weights from test 2)
6	Fine-tuning	MultiSenGE	Pretrained (MultiSenNA)	
7	Encoder Freezing	MultiSenNA	Pretrained (MultiSenGE)	Freezing encoder, training on MultiSenNA
8	Encoder Freezing	MultiSenGE	Pretrained (MultiSenNA)	Freezing encoder, training on MultiSenGE
9	Reduced Dataset	MultiSenNA	Pretrained (MultiSenGE)	Training and validation set size reduced by 4×
10	Reduced Dataset	MultiSenGE	Pretrained (MultiSenNA)	Training and validation set size reduced by 4×

Table 2. Results of Ensemble Transfer Learning for MultiSenNA and MultiSenGE (in bold the best metrics and for carbon impact and time, lower is better)

Test	MultiSenNA				MultiSenGE			
	Accuracy (%)	F1-Score (%)	Carbon (kg)	Time (h)	Accuracy (%)	F1-Score (%)	Carbon (kg)	Time (h)
1	79.15	81.30	0.6352	30.04	-	-	-	-
2	81.75	83.09	0.4114	18.31	-	-	-	-
3	-	-	-	-	81.61	81.74	0.3997	18.62
4	-	-	-	-	83.07	84.66	0.4594	22.24
5	81.90	83.17	0.4623	21.19	-	-	-	-
6	-	-	-	-	85.55	86.47	0.2868	12.85
7	75.35	78.09	0.7838	38.55	-	-	-	-
8	-	-	-	-	82.38	84.16	0.5571	27.59
9	79.79	81.24	0.1157	5.28	-	-	-	-
10	-	-	-	-	82.97	84.63	0.0723	3.25

fine-tuning was applied with pretrained weights from MultiSenNA. This highlights the reciprocal benefits of transfer learning between the two models.

In addition to the accuracy and weighted F1-Score analyses, an important aspect of this study was the assessment of the carbon impact associated with the transfer learning approaches. This evaluation is needed in the current context, where the environmental footprint of AI models is exponentially increasing, especially for large-scale applications such as LULC classification. In the case of MultiSenNA, the most notable reduction in carbon impact was recorded in test 9 (0.1157), which utilized a reduced dataset size. This experiments, while significantly minimizing the carbon footprint, exhibited only a modest drop in the weighted F1-Score compared to the full dataset test (test 5 with 83.17% vs. test 9 with 81.24%). Similarly, for MultiSenGE, test 10, which also used a smaller dataset, recorded the lowest carbon footprint (0.0723). Similarly, the performance measured by the weighted F1-score remained relatively close to that observed on the full dataset test (test 6 with 86.47% vs. test 10 with 84.63%). These findings demonstrate the importance of considering environmental impacts in the development and deployment of AI models for geographic and remote sensing applications and shows that we can consider a smaller dataset size coupled with transfer learning (e.g. fine-tuning) approaches. Furthermore, the results highlight the need for a holistic approach in model evaluation, considering not only accuracy and performance metrics but also the environmental costs associated with model training and inference, especially in a context of global change.

If we focus the comparison of F1-Score only on classes 1 to 5 related to urban fabrics (we computed the weighted F1-Score for classes 1 to 5, excluding the others), a comparison of weighted F1-Scores between the full and small datasets for both Multi-

Table 3. Comparison of Weighted F1-Scores for Urban Fabric Analysis.

Dataset	Full dataset (%)	Small dataset (%)
MultiSenGE	60.44	55.85
MultiSenNA	53.54	49.82

SenGE and MultiSenNA was carried out, as shown in Table 3. The weighted F1-Score presented differences between the two approaches. For the MultiSenGE, the weighted F1-Score was 60.44% for the full dataset and 55.85% for the small dataset. Conversely, MultiSenNA achieved a weighted F1-Score of 53.54% on the full dataset and 49.82% on the small dataset. This indicates a relatively stronger performance of MultiSenGE in both dataset scenarios, probably due to less complex urban areas. Indeed, in the Grand-East region the urban classes are more compact due to the planification rules in comparison on the NA region where the housing is more fragmented. The difference between both approaches is more significant for the urban fabric classes than the entire LULC classes.

6. Conclusion

This research introduces the MultiSenNA dataset and AI4LCC collection, a novel multitemporal and multimodal semantic segmentation and classification dataset covering the *Nouvelle-Aquitaine* region of France. This dataset opens new research pathways in transfer learning using time series remote sensing imagery. Our study investigates the application of fine-tuning pre-trained networks on the MultiSenNA dataset, building upon the well-established hypothesis that fine-tuning is often superior to training from scratch. By applying this paradigm to our dataset, we aim to provide new insights into its behavior in a multimodal and multitemporal context. The experiments reveal that fine-tuning delivers robust performance on both full and small datasets, underscoring its effectiveness and scalability. A key finding is the comparable efficacy of fine-tuning on reduced dataset sizes, maintaining high model performance while significantly lowering the carbon footprint and the execution time.

In summary, the creation of the MultiSenNA dataset and the implementation of fine-tuning techniques represent an advancement in crafting flexible and sustainable AI models for a wide array of uses. This research underscores the importance of further exploring foundation models (Jakubik et al. 2023; Lu et al. 2024) customized for different datasets, leading towards more refined and sustainable developments in AI, particularly in the domain of remote sensing. The entire processing chain in remote sensing, from data cubes acquisition to preprocessing and model training, consumes substantial amounts of energy. Optimizing these processes, especially in the context of large-scale land use and land cover mapping, is essential to mitigate the environmental impact of such computationally intensive workflows.

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Data Availability Statement

MultiSenGE and MultiSenNA are freely available through the AI4LCC collection website: <https://doi.theia.data-terra.org/ai4lcc/?lang=en>

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